

Spectral Methods for the 2D Schrödinger Equation

Exercises 1–6

1 Exercise 1 — Eigenvalues of the Spectral Laplacian

The 2D spectral Laplacian on the unit square $(0, 1)^2$ with zero Dirichlet boundary conditions is exactly diagonalised by the tensor-product sine matrix $S = S_x \otimes S_y$. The eigenvalues are

$$\Gamma_{pq} = -(p\pi)^2 - (q\pi)^2, \quad p = 1, \dots, N_x - 1, \quad q = 1, \dots, N_y - 1.$$

The implementation uses broadcasting over a column vector p and a row vector q to fill the $(N_x - 1) \times (N_y - 1)$ matrix without any explicit loops:

```
1 function make_eigenvalues(Nx::Int, Ny::Int)
2     p = (1:Nx-1) .* pi
3     q = (1:Ny-1)' .* pi
4     Gamma = @. -(p^2) - (q^2)
5     return Gamma
6 end
```

2 Exercise 2 — Crank–Nicolson Time Step

The Crank–Nicolson scheme for $i \partial_t \psi = -\Delta \psi$ is

$$\left(I - \frac{i\Delta t}{2} \Delta_h\right) \psi^{n+1} = \left(I + \frac{i\Delta t}{2} \Delta_h\right) \psi^n.$$

Applying the 2D DST to both sides and using Proposition 3.1 from the notes, each spectral mode decouples:

$$\hat{U}_{pq}^{n+1} = \rho_{pq} \hat{U}_{pq}^n, \quad \rho_{pq} = \frac{1 + \frac{i\Delta t}{2} \Gamma_{pq}}{1 - \frac{i\Delta t}{2} \Gamma_{pq}}.$$

Since $\Gamma_{pq} < 0$, the numerator and denominator are complex conjugates so $|\rho_{pq}| = 1$ exactly — the scheme preserves the discrete L^2 norm for any Δt .

The algorithm is three operations per step:

1. Forward DST: $\hat{U} \leftarrow \text{dst2d}(U)$.
2. Pointwise multiply: $\hat{U} \leftarrow \rho \odot \hat{U}$.
3. Inverse DST: $U_{\text{new}} \leftarrow \text{idst2d}(\hat{U})$ (same call as dst2d since $S = S^{-1}$).

```
1 function cn_step(U, Gamma, dt::Real, Nx::Int, Ny::Int)
2     Uhat = dst2d(U, Nx, Ny)
3     rho = (1 .+ im*dt/2 .* Gamma) ./ (1 .- im*dt/2 .* Gamma)
4     U_new = rho .* Uhat
5     return idst2d(U_new, Nx, Ny)
6 end
```

3 Exercise 3 — Simulation Loop

The simulation loop runs `cn_step` for `nsteps` iterations and saves a snapshot every `save_every` steps together with the corresponding time. The initial state is cast to `ComplexF64` at entry so that the snapshot vector stays complex even when `psi0` is real-valued.

```

1 function simulate(psi0, Gamma, dt::Real, nsteps::Int, Nx::Int, Ny::Int;
2                 save_every::Int = 1)
3     psi      = complex(copy(psi0))
4     snapshots = [copy(psi)]
5     times     = [0.0]
6     for step in 1:nsteps
7         psi = cn_step(psi, Gamma, dt, Nx, Ny)
8         if step % save_every == 0
9             push!(snapshots, copy(psi))
10            push!(times, step * dt)
11        end
12    end
13    return snapshots, times
14 end

```

4 Exercise 4 — Laplacian and Norm Verification

(a) Eigenvalue Check

We sample $u(\mathbf{x}) = \sin(\pi x) \sin(2\pi y)$ on an 8×8 interior grid. This function is an eigenfunction of $-\Delta$ with eigenvalue $\lambda = (1^2 + 2^2)\pi^2 = 5\pi^2$, so we expect

$$\Delta_h u = -5\pi^2 u$$

at every interior node.

```

1 Nx, Ny = 8, 8
2 x = (1:Nx-1) ./ Nx
3 y = (1:Ny-1) ./ Ny
4 U      = [sin(pi*xi)*sin(2*pi*yi) for xi in x, yi in y]
5 Gamma  = make_eigenvalues(Nx, Ny)
6 LU     = apply_laplacian(U, Gamma, Nx, Ny)
7 expected = -5*pi^2 .* U
8 println(maximum(abs.(LU .- expected)))

```

Result: the maximum pointwise error is 4.26×10^{-14} , which is at machine-epsilon level. This confirms that `apply_laplacian` implements the spectral Laplacian exactly.

(b) Norm Conservation

We run the Gaussian wave-packet simulation

$$\psi_0(\mathbf{x}) = C \exp\left(-\frac{|\mathbf{x} - \mathbf{x}_0|^2}{2\sigma^2}\right) \exp(i\mathbf{k} \cdot \mathbf{x}), \quad \mathbf{x}_0 = (0.3, 0.3), \sigma = 0.04, \mathbf{k} = (15\pi, 7\pi),$$

on a 128×128 interior grid for 2000 steps with $\Delta t = 5 \times 10^{-4}$ and record the discrete L^2 norm at every step.

Result: the maximum drift in $\|\psi^n\|_{L^2}$ is 4.02×10^{-13} , consistent with floating-point round-off. The norm is preserved to machine precision for all 2000 steps, which matches the theoretical guarantee $|\rho_{pq}| = 1$.

5 Exercise 5 — Eigenstate Accuracy

We choose $(p_0, q_0) = (2, 3)$ and initialise with the eigenstate

$$\psi_0(\mathbf{x}) = C \sin(2\pi x) \sin(3\pi y),$$

whose exact solution is

$$\psi(\mathbf{x}, t) = \psi_0(\mathbf{x}) e^{-i\lambda t}, \quad \lambda = -(2\pi)^2 - (3\pi)^2 \approx -128.3.$$

(a) Density Conservation

Since $|\psi(\mathbf{x}, t)|^2 = |\psi_0(\mathbf{x})|^2 |e^{-i\lambda t}|^2 = |\psi_0(\mathbf{x})|^2$, the probability density must not change over time.

Result: the maximum drift in $|\psi^n|^2$ over 200 steps with $\Delta t = 5 \times 10^{-4}$ is 3.59×10^{-13} , which confirms density conservation to machine precision.

(b) Phase Error Scaling

The numerical phase at step n is $\theta^n = \arg\langle\psi_0, \psi^n\rangle$ where $\langle u, v \rangle = h_x h_y \sum_{j,k} \bar{u}_{jk} v_{jk}$. The exact phase is $-\lambda_{p_0 q_0} n \Delta t$. Since $\lambda < 0$ in our sign convention, the exact phase equals λt (negative).

We restrict to $T = 0.01$ so that $|\lambda T| \approx 1.28 < \pi$, avoiding phase wrap-around in `angle()`.

The Crank–Nicolson amplification factor satisfies

$$\arg(\rho_{pq}) = 2 \arctan\left(\frac{\lambda \Delta t}{2}\right) = \lambda \Delta t - \frac{(\lambda \Delta t)^3}{12} + O(\Delta t^5),$$

so the phase error per step is $O(\Delta t^3)$ and after $n = T/\Delta t$ steps the accumulated error is $O(\Delta t^2)$.

Δt	max phase error	rate
10^{-2}	1.42×10^{-1}	—
5×10^{-3}	4.15×10^{-2}	1.78
10^{-3}	1.76×10^{-3}	1.96
5×10^{-4}	4.40×10^{-4}	2.00

The rate approaches 2 as $\Delta t \rightarrow 0$, confirming second-order accuracy in the accumulated phase error.

6 Exercise 6 — Second-Order Time Accuracy

Setup

We fix $T = 0.05$ and use the §5.1 Gaussian initial condition on a 128×128 grid. The reference solution is computed with $\Delta t_{\text{ref}} = 10^{-5}$ (5000 steps). For $\Delta t \in \{0.02, 0.01, 0.005, 0.0025\}$ we measure $\|U(\cdot, T) - U_{\text{ref}}(\cdot, T)\|_F$.

Results

Δt	$\ U - U_{\text{ref}}\ _F$	rate
0.02	186.1	—
0.01	177.4	0.07
0.005	173.3	0.03
0.0025	182.5	−0.08

Discussion

The errors do not decrease and the expected slope of 2 is not observed. To understand why, we check how much the solution changes in a single step:

$$\|\psi^0\|_F = 128, \quad \|\psi^1 - \psi^0\|_F \approx 255 \quad (\Delta t = 0.02).$$

The change in one step is already larger than the norm of the initial condition, which means $\Delta t = 0.02$ is far outside the asymptotic regime for this initial condition.

The reason is that the Gaussian ψ_0 with $\sigma = 0.04$ is very narrow and contains high-frequency spectral modes. The highest modes have eigenvalues of order $\lambda_{\max} \sim -(N\pi)^2$. The Crank–Nicolson phase rotation per step for such a mode is $|\lambda_{\max}|\Delta t \gg \pi$ at $\Delta t = 0.02$, so the mode is severely under-resolved in time even though the scheme is unconditionally stable.

Although Crank–Nicolson preserves the norm for any Δt , achieving second-order accuracy in $\|U_{\Delta t} - U_{\text{ref}}\|_F$ requires Δt to be in the asymptotic regime where the phase error is small. The prescribed Δt values do not satisfy this condition for the §5.1 initial condition.

As a sanity check, second-order convergence *is* confirmed in Exercise 5(b) using an eigenstate with a single frequency $\lambda \approx -128.3$, where the prescribed Δt values satisfy $|\lambda|\Delta t \leq 1.28 < \pi$ and the asymptotic regime is reached.